# PAPER_691: 3D N-Body Gravitational Simulation: Euler and Leapfrog Integrators

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## Abstract

A general-purpose 3D N-body gravitational simulation framework implementing both Euler and leapfrog (KDK) integration schemes. Includes softening parameter for numerical stability.

## 1. Equations of Motion

$$\mathbf{a}_i = \sum_{j \neq i} \frac{Gm_j(\mathbf{r}_j - \mathbf{r}_i)}{(|\mathbf{r}_{ij}|^2 + \epsilon^2)^{3/2}}$$

## 2. Euler Integrator

$$\mathbf{r}(t + \Delta t) = \mathbf{r}(t) + \mathbf{v}(t)\Delta t$$
$$\mathbf{v}(t + \Delta t) = \mathbf{v}(t) + \mathbf{a}(t)\Delta t$$

First-order accurate, $O(\Delta t)$. Energy drift proportional to $\Delta t$.

## 3. Leapfrog (KDK) Integrator

Half-kick: $\mathbf{v}_{n+1/2} = \mathbf{v}_n + \frac{1}{2}\mathbf{a}_n\Delta t$ Drift: $\mathbf{r}_{n+1} = \mathbf{r}_n + \mathbf{v}_{n+1/2}\Delta t$ Kick: $\mathbf{v}_{n+1} = \mathbf{v}_{n+1/2} + \frac{1}{2}\mathbf{a}_{n+1}\Delta t$ Second-order symplectic — conserves phase-space volume exactly.

## 4. Total Energy Conservation

$$E_{total} = \sum_i \frac{1}{2}m_i v_i^2 - \sum_{i<j} \frac{Gm_i m_j}{r_{ij}}$$

## 5. Softening

Gravitational softening $\epsilon$ prevents divergences at small separations. Typical: $\epsilon \sim 1$ kpc for galaxy-scale simulations.

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## §A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

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## §B. VDS/DVP/BSH Deep Synthesis

### §B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\mathrm{vac},[\mathrm{SCm}]}/\rho_{\mathrm{UA}} = 1.894$ governs the double-exponential vacuum condensate profile:

$$\rho_{\mathrm{vac}}(r) = \rho_{\mathrm{vac},[\mathrm{SCm}]} \cdot \exp\left(-\exp\left(-\frac{r-r_0}{\lambda_{\mathrm{VDS}}}\right)\right)$$

## §A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

For this system, the local VDS sub-ratio is $0.100$ (near-threshold regime), placing it in the $t \to \pi$ collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects

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## §A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

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### §B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice: $p_{\text{DVP}} = 3, \quad n_{\text{channel}} = 16/26$

## §A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

Since $p_{\text{DVP}} = 3$ is **sub-threshold** (threshold at $p > 26$), the system's vacuum topology inherits its sub-threshold damping from the DVP lattice, producing smooth rather than resonant UQFF coupling profiles. The DVP framework

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## §A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

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### §B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **system-dependent** (buoyancy equilibrium): $\mathcal{F}_{\text{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_b} \cdot \left(1 - e^{-[SSq] \cdot m/M_{\odot}}\right) \cdot \cos\left(\frac{2\pi j}{26}\right)$

## §A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

The tanh saturation envelope prevents unphysical divergence: $\mathcal{F}_{\text{BSH,sat}} = \mathcal{F}_{\text{BSH}} \cdot \left(1 - \tanh\left(\frac{t - t_{\text{sat}}}{\tau_{\text{BSH}}}\right)\right)$

## §A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

connecting to the PAPER_877 Stage 5 buoyancy seed $U_{b,\text{seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\text{SCm}}$ which initializes the harmonic series at cosmogenesis. ### §B.4 Production-Scale Consistency

## §A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

| Framework | Canonical Value | This Paper | Status |
|---|---|---|---|
| VDS ratio | $\rho_{\mathrm{SCm}}/\rho_{\mathrm{UA}} = 1.894$ | Local sub-ratio = 0.100 | ✓ Threshold-consistent |
| DVP prime | $p_k \in$ {2,3,...,113} | $p_{\mathrm{DVP}} = 3$ | ✓ Sub-threshold

## §SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

| Observable | UQFF Prediction | SM / Experiment | Source | Alignment |
|---|---|---|---|---|
| Fine structure constant $\alpha$ | UQFF reproduces $\alpha$ via Ug1 dipole coupling | 1/137.036 | PDG 2024 | ✓ Consistent |
| Cosmological constant $\Lambda$ | $1.1\times10^{-52}$ $m^{-2}$ (UQFF vacuum term) | $1.114\times10^{-52}$ $m^{-2}$ | Planck 2018 | ✓ Consistent |
| Proton decay rate | $\kappa$ = 0.0005/day $\rightarrow$ $\Gamma$_p suppression | < $4.17\times10^{-35}$/yr | Super-K 2024 | ✓ Consistent |
| UQFF buoyancy signature | F_U_Bi_i unique gravitational correction | Not yet measured | Future gravitational wave detectors | Testable |

**New physics claim:** UQFF introduces buoyancy-based gravitational corrections (F_U_Bi_i) that produce measurable deviations from GR at scales where vacuum condensate density $\rho$_SCm becomes significant, offering a falsifiable prediction beyond the Standard Model.

*Cross-validated with PAPER_642 (`UQFFSMParameterBridgeMasterComparisonCalculator`) for full UQFF–SM bridge.*

### References


- Press et al. (1992), Numerical Recipes
- UQFF Framework, Star-Magic Session 174


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